

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

	Level/CO	Marks
Q. 1 Objective type questions. (Compulsory Question)		12
1 smallest change in a measured variable to which an instrument will respond is:		1
a) Accuracy b) Resolution c) Precision d) drift		
2 The difference between the indicated value and the true value of a quantity is :		1
a) gross error b) dynamic error c) absolute error d) relative error		
3 If the instrument has square law response, it can be used for the instrument of:		1
a) DC voltage only b) AC voltage & current c) Both AC and DC quantities d) Resistance only		
4 The primary current in CT is detected by:		1
a) Secondary burden b) core of transformer c) load current d) none of the above		
5 The Wheatstone bridge method of resistance measurement is ideally suitable for the Measurement		1
a) 0.001Ω to 1Ω b) 0.1Ω to 100Ω c) 100Ω to $10 K\Omega$ d) $100 k\Omega$ to $10 M\Omega$.		
6 Schering bridge can be used to measure which one of the followings:		1
a) Inductance b) Resistance c) Capacitance d) Power		
7 Which bridge measures self-inductance using a standard capacitor?		1
a) Maxwell bridge b) Schering bridge c) Anderson bridge d) Wien bridge		

8	A digital storage oscilloscope (DSO) is used to:			1
	a) Generate sinusoidal signals	b) Measure instantaneous power	c) Store and analyze waveforms digitally	d) Measure electrical resistance
9	Which transducer is commonly used for temperature measurement?			1
	a) RVDT	b) Thermocouple	c) LVDT	d) Strain gauge
10	Which of the following is NOT a type of transducer?			1
	a) Thermistor	b) LVDT	c) Hall effect sensor	d) Oscilloscope
11	Which characteristic is crucial for selecting a transducer?			1
	a) Durability	b) Sensitivity	c) Size	d) Cost
12	Signal conditioning is essential for:			1
	a) Direct data transmission	b) Converting raw signals into a usable format	c) Measuring mechanical stress	d) Storing large amounts of data
Q. 2 Solve the following.				
	A) Define error. Explain the different types of errors with example.		CO1	6
	B) Describe the characteristics of instruments & measurement system.		CO2	6
Q. 3 Solve the following.				
	A) Draw a neat sketch of PMMC instrument and derive the equation and explain in detail?		CO1	6
	B) Explain two wattmeter method for measurement of power in three phase circuit.		CO2	6
Q. 4 Solve Any Two of the following.				
	A) The inductance of moving iron ammeter is given by following expression $L = (20 + 10\theta - 2\theta^2) \mu\text{H}$, where, θ is deflection in radians. The spring constant is $24 \times 10^{-6} \text{ Nm/rad}$. Calculate the values of deflection for a current 5A.		CO1	6
	B) With a neat, labeled diagram explain Maxwell's Inductance bridge and obtain the output equation for L_x and R_x		CO2	6
	C) With a circuit & phasor diagram derive the equation for an unknown self-inductance measurement using HAY'S bridge.		CO3	6
Q. 5 Solve Any Two of the following.				
	A) Explain the working of Q Meter.		CO1	6
	B) Draw and explain Ramp type DVM in detail.		CO2	6
	C) Explain in detail the construction and working of digital storage oscilloscope.		CO3	6

Q. 6 Solve Any Two of the following.

12

A) Explain the working of strain gauge in detail.

CO1

6

B) Explain different characteristics of transducer.

CO2

6

C) With a neat, labeled diagram explain the operation of LVDT.

CO3

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